CAT 2025 Slot 1 QA Question Paper with Solutions

Time Allowed :40 Minutes	**Maximum Marks :**66	**Total Questions :**22

General Instructions

Read the following instructions very carefully:

1. The total duration of the QA section is **40 minutes**.
2. Each correct answer carries **+3 marks**.
3. For multiple-choice questions (MCQs), **−1 mark** will be deducted for each wrong answer.
4. There is **no negative marking** for Type-in-the-Answer (TITA) questions.
5. Once the allotted time for a section ends, you will automatically move to the next section. You cannot go back to a previous section.
6. All answers must be entered on the computer screen using the mouse and keyboard. Rough sheets will be provided.
7. The use of calculators, mobile phones, or any other electronic device is strictly prohibited. An on-screen basic calculator will be provided.

1. A 200-litre container holds a solution that is 30% acid and the rest water. The solution undergoes the following three processes sequentially:
• 20% of the water content is evaporated. From the remaining mixture, 10% of the acid content is chemically extracted and removed. Finally, 15% of the resulting solution is removed and replaced with water.
What is the volume of acid in the final solution?

Correct Answer: 45.9 Litres

Solution:
Step 1: Calculate the initial volume of acid and water.

Total volume of the solution = 200 Litres.

Concentration of acid = 30%.

Initial Volume of Acid $= 200 \times \frac{30}{100} = 60$ Litres.

Initial Volume of Water $= 200 - 60 = 140$ Litres.

Step 2: Process 1 - Evaporation of water.

20% of the water content is evaporated.

Volume of water evaporated $= 140 \times \frac{20}{100} = 28$ Litres.

Remaining Volume of Water $= 140 - 28 = 112$ Litres.

The volume of acid remains unchanged at 60 Litres.

Step 3: Process 2 - Extraction of acid.

10% of the acid content is chemically extracted and removed from the remaining mixture.

Volume of acid extracted $= 60 \times \frac{10}{100} = 6$ Litres.

Remaining Volume of Acid $= 60 - 6 = 54$ Litres.

The volume of water remains unchanged at 112 Litres.

Step 4: Process 3 - Removal and replacement.

15% of the resulting solution is removed.

Since the solution is a homogeneous mixture, removing 15% of the solution removes 15% of the acid present in it.

Volume of acid removed $= 54 \times \frac{15}{100} = 54 \times 0.15 = 8.1$ Litres.

Remaining Volume of Acid $= 54 - 8.1 = 45.9$ Litres.

The subsequent step of replacing the removed solution with water adds only water and does not affect the volume of acid.

Conclusion:

The final volume of acid in the solution is 45.9 Litres.

Quick Tip

When calculating changes in a mixture, treat each component (acid and water) separately when the process affects only one of them (like evaporation or specific extraction). When a portion of the total mixture is removed, reduce the quantity of all components by the same percentage.

2. In the sequence 1, 3, 5, 7, ..., k, ..., 57, the sum of the numbers up to k, excluding k, is equal to the sum of the numbers from k up to 57, also excluding k. What is k?

Correct Answer: 41

Solution:
Step 1: Analyze the sequence.

The sequence $1, 3, 5, 7, \ldots, 57$ is an Arithmetic Progression consisting of consecutive odd numbers.

First term $(a) = 1$, Common difference $(d) = 2$.

Let N be the total number of terms. The last term is 57.

$a_N = 1 + (N-1)2 = 57 \implies 2(N-1) = 56 \implies N = 29.$

Step 2: Define expressions for the sums relative to k.

Let k be the m^{th} term of the sequence. So, $k = 2m - 1$.

The sum of the first n odd numbers is equal to n^2.

The "sum of numbers up to k, excluding k" corresponds to the sum of the first $m-1$ terms.

$\text{Sum}_{before} = (m-1)^2$.

The "sum of numbers from k up to 57, excluding k" corresponds to the sum of the remaining terms after the m^{th} term (from $m+1$ to 29).

$\text{Sum}_{after} = (\text{Sum of first 29 terms}) - (\text{Sum of first } m \text{ terms})$.

$\text{Sum}_{after} = 29^2 - m^2$.

Step 3: Solve for m.

Equate the two sums as per the problem statement:

$(m-1)^2 = 29^2 - m^2$

$m^2 - 2m + 1 = 841 - m^2$

$2m^2 - 2m + 1 - 841 = 0$

$2m^2 - 2m - 840 = 0$

Divide the equation by 2:

$m^2 - m - 420 = 0$

Factorize the quadratic equation:

$(m - 21)(m + 20) = 0$

Since the position m must be positive, $m = 21$.

Step 4: Find the value of k.

Calculate the 21st term:

$k = 2(21) - 1 = 42 - 1 = 41.$

> Quick Tip
>
> For any sequence of consecutive odd numbers starting from 1 $(1, 3, 5, \dots)$, the sum of the first n terms is always n^2. Using this property avoids the longer arithmetic progression sum formula.

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3. Find the number of real values of x satisfying the equation:

$$\log_2(x^2 - 5x + 6) + \log_{1/2}(x - 2) = 3$$

Correct Answer: 1

Solution:
Step 1: Determine the valid domain for x.

For the logarithm functions to be defined, their arguments must be strictly positive:

Condition 1: $x^2 - 5x + 6 > 0$

Factorizing the quadratic: $(x - 2)(x - 3) > 0$

This implies $x < 2$ or $x > 3$.

Condition 2: $x - 2 > 0 \implies x > 2$.

Taking the intersection of both conditions, the valid domain is $x > 3$.

Step 2: Simplify the equation using logarithm properties.

Recall the base change property: $\log_{1/a}(b) = -\log_a(b)$.

So, $\log_{1/2}(x-2) = \log_{2^{-1}}(x-2) = -\log_2(x-2)$.

Substitute this back into the original equation:

$\log_2(x^2 - 5x + 6) - \log_2(x-2) = 3$.

Step 3: Combine the terms.

Use the quotient rule $\log_a(M) - \log_a(N) = \log_a(\frac{M}{N})$:

$\log_2\left(\frac{x^2-5x+6}{x-2}\right) = 3$.

Substitute the factors from Step 1:

$\log_2\left(\frac{(x-2)(x-3)}{x-2}\right) = 3$.

Since $x > 3$, $x \neq 2$, so we can cancel $(x-2)$:

$\log_2(x-3) = 3$.

Step 4: Solve for x.

Convert the logarithmic equation to exponential form:

$x - 3 = 2^3$

$x - 3 = 8$

$x = 11$.

Step 5: Validate the solution.

Check if $x = 11$ lies in the domain defined in Step 1 ($x > 3$).

Since $11 > 3$, the solution is valid.

Therefore, there is exactly one real value of x.

💡 Quick Tip

Always define the domain of logarithmic functions (argument ¿ 0) before simplifying or solving the equation. This prevents accepting extraneous roots that result from algebraic manipulations.

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4. Find the number of integer pairs (x, y) that satisfy the following system of inequalities:

$$\begin{cases} x \geq y \geq 3 \\ x + y \leq 14 \end{cases}$$

Correct Answer: 25

Solution:
Step 1: Determine the constraints on y.

From the given inequalities, we have $x \geq y$. Substituting this into the second inequality:

$y + y \leq x + y \leq 14 \implies 2y \leq 14 \implies y \leq 7.$

We are also given that $y \geq 3$. Therefore, the possible integer values for y are $3, 4, 5, 6, 7$.

Step 2: Determine the range of x for each valid y.

For a fixed value of y, x must satisfy $y \leq x$ and $x \leq 14 - y$. Thus, $x \in [y, 14 - y]$.

The number of integer values for x is $(14 - y) - y + 1 = 15 - 2y$.

Step 3: Calculate the number of pairs for each case.

Case 1: $y = 3$

Number of x values $= 15 - 2(3) = 15 - 6 = 9$. (Pairs: $(3, 3), (4, 3), \ldots, (11, 3)$)

Case 2: $y = 4$

Number of x values $= 15 - 2(4) = 15 - 8 = 7$.

Case 3: $y = 5$

Number of x values $= 15 - 2(5) = 15 - 10 = 5$.

Case 4: $y = 6$

Number of x values $= 15 - 2(6) = 15 - 12 = 3$.

Case 5: $y = 7$

Number of x values $= 15 - 2(7) = 15 - 14 = 1$. (Pair: $(7, 7)$)

Step 4: Sum the results.

Total pairs $= 9 + 7 + 5 + 3 + 1 = 25$.

> Quick Tip
>
> To count integer solutions for a system of inequalities, fix the variable with the most restricted range (in this case, y) and iterate through its possible values to find the count of the second variable.

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5. Given that a, b, and c are real numbers satisfying the equations:

1. $2a - 5b + 11c = 0$
2. $11a + 10b - 2c = 5$

Find the value of the expression $(a^2 - b^2 + c^2)$.

(A) 5/11
(B) 2/13
(C) 1
(D) CBD (Cannot be determined)

Correct Answer: (D) CBD (Cannot be determined)

Solution:
Step 1: Express variables a and b in terms of c.

We have a system of two linear equations with three variables (a, b, c). This means the system is underdetermined, and the solutions for a and b will depend on the parameter c.

Equation (1): $2a - 5b + 11c = 0$

Equation (2): $11a + 10b - 2c = 5$

To eliminate b, multiply Equation (1) by 2:

$4a - 10b + 22c = 0$

Add this to Equation (2):

$(4a + 11a) + (-10b + 10b) + (22c - 2c) = 0 + 5$

$15a + 20c = 5$

Divide by 5: $3a + 4c = 1 \implies 3a = 1 - 4c \implies a = \frac{1-4c}{3}$.

Step 2: Solve for b in terms of c.

From Equation (1), $5b = 2a + 11c$. Substitute the value of a:

$5b = 2\left(\frac{1-4c}{3}\right) + 11c$

$5b = \frac{2-8c}{3} + \frac{33c}{3}$

$5b = \frac{2+25c}{3}$

Divide by 5: $b = \frac{2+25c}{15}$.

Step 3: Substitute a and b into the target expression.

Let $E = a^2 - b^2 + c^2$.

Substitute $a = \frac{5-20c}{15}$ (scaling by 5/5 for a common denominator) and $b = \frac{2+25c}{15}$:

$E = \left(\frac{5-20c}{15}\right)^2 - \left(\frac{2+25c}{15}\right)^2 + c^2$

Factor out $\frac{1}{225}$:

$E = \frac{1}{225}\left[(5-20c)^2 - (2+25c)^2 + 225c^2\right]$

Step 4: Expand and simplify the terms.

$(5 - 20c)^2 = 25 - 200c + 400c^2$

$(2 + 25c)^2 = 4 + 100c + 625c^2$

Subtracting the second from the first:

$(25 - 4) + (-200c - 100c) + (400c^2 - 625c^2) = 21 - 300c - 225c^2$

Now add the $225c^2$ term from the original expression:

Numerator $= (21 - 300c - 225c^2) + 225c^2 = 21 - 300c$

Thus, $E = \frac{21-300c}{225} = \frac{7-100c}{75}$.

Conclusion:

The value of the expression depends linearly on c. Since c can be any real number, the value of the expression is not unique.

Therefore, the value cannot be determined.

> Quick Tip
>
> In systems with fewer equations than variables (underdetermined systems), the solution usually depends on a free parameter. If the target expression does not cancel out this parameter perfectly, the value is not unique, leading to a "Cannot be determined" answer.

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6. A shopkeeper offers a 22% discount on the marked price of chairs. He gives 13 chairs to a customer at the discounted price of 12 chairs. If he still makes a profit of 26%, what is the marked price (MP) of a single chair, assuming its cost price (CP) is Rupees 100?

Correct Answer: (B) Rs. 175

Solution:
Step 1: Calculate the Total Cost Price (CP).

The shopkeeper parts with 13 chairs in this transaction.

Given CP of 1 chair = Rs. 100.

Total CP for 13 chairs $= 13 \times 100 =$ Rs. 1300.

Step 2: Calculate the Total Selling Price (SP) required.

The shopkeeper makes a profit of 26% on the total cost.

Total SP = Total CP $\times(1 + \frac{\text{Profit\%}}{100})$.

Total SP $= 1300 \times 1.26 =$ Rs. 1638.

Step 3: Formulate the Selling Price in terms of Marked Price (MP).

Let the Marked Price of one chair be MP.

The discount offered is 22%, so the discounted price of one chair is:

$\text{Price}_{discounted} = MP \times (1 - 0.22) = 0.78MP$.

The deal states that the customer receives 13 chairs but pays the price of only 12 chairs at the discounted rate.

Total SP $= 12 \times (0.78MP) = 9.36MP$.

Step 4: Equate the Selling Prices and solve for MP.

$9.36MP = 1638$

$MP = \frac{1638}{9.36}$

$MP = \frac{163800}{936}$

$MP = 175$.

Conclusion:

The Marked Price of a single chair is Rs. 175.

> Quick Tip
>
> In "Buy x Get y" or similar quantity-based deals, always equate the Total Cost of the actual inventory leaving the shop (y items) against the Total Revenue received for the paid items (x items).

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7. In a class, there are 10 boys and 'n' girls. After some students leave, 60% of the boys and 40% of the girls remain. If the number of remaining girls is 8 more than the number of remaining boys, what was the initial number of girls, n?

(A) 20

(B) 25

(C) 30

(D) 35

Correct Answer: (D) 35

Solution:
Step 1: Calculate the number of remaining boys.

Initial number of boys = 10.

Percentage of boys remaining = 60%.

Remaining Boys = $10 \times \frac{60}{100} = 6$.

Step 2: Formulate the equation for remaining girls.

Initial number of girls = n.

Percentage of girls remaining = 40%.

Remaining Girls = $n \times \frac{40}{100} = 0.4n$.

Step 3: Apply the given condition to find n.

The problem states that the number of remaining girls is 8 more than the number of remaining boys.

Remaining Girls = Remaining Boys + 8

$0.4n = 6 + 8$

$0.4n = 14$

$n = \frac{14}{0.4} = \frac{140}{4}$

$n = 35$.

Conclusion:

The initial number of girls was 35.

💡 Quick Tip

Break the problem into "Initial State" and "Final State". Translate the percentage changes into absolute numbers where possible (like for the boys here) to simplify the equation for the unknown variable.

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8. Stocks A, B and C, at 120, 90 and 80 rs respectively. A trader holds a portfolio consisting 60 shares of A, 20 of B and C together. If total value is 33000 what is the number of shares he holds.
(A) 320
(B) 360
(C) 380
(D) 400
Correct Answer: (C) 380

Solution:
Step 1: Calculate the value of the shares of Stock A.

Number of shares of A = 60.

Price of A = Rs. 120.

Value of A = $60 \times 120 =$ Rs. 7200.

Step 2: Interpret the holdings of Stock B.

The phrase "20 of B and C together" in the context of the high total value implies the trader has "20 shares of B" and an unknown number of shares of C. (Any other interpretation, such as sum of B and C being 20, yields a value far too low compared to Rs. 33000).

Number of shares of B = 20.

Price of B = Rs. 90.

Value of B = $20 \times 90 =$ Rs. 1800.

Step 3: Calculate the value and number of shares of Stock C.

Total Value of Portfolio = Rs. 33000.

Value of C = Total Value - (Value of A + Value of B).

Value of C = $33000 - (7200 + 1800) = 33000 - 9000 =$ Rs. 24000.

Price of C = Rs. 80.

Number of shares of C = $\frac{24000}{80} = 300$ shares.

Step 4: Calculate the total number of shares.

Total Shares = Shares of A + Shares of B + Shares of C.

Total Shares $= 60 + 20 + 300 = 380$.

Conclusion:

The trader holds a total of 380 shares.

Quick Tip

In portfolio problems with ambiguous wording, use the total value constraint to validate your interpretation of the quantities. If a literal interpretation (like "sum of B and C is 20") contradicts the financial totals, look for the interpretation that yields valid integer counts for all stocks.

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9. Arun, Tarun and Varun work for 24, 21 and 15 days respectively and get paid 2160, 2400 and 2160 rupees respectively. They get paid the same even if they work for a partial day. If the work has to be completed within 10 days or less, what is the minimum amount that has to be paid to complete the entire task?

(A) Rs. 2100

(B) Rs. 2160

(C) Rs. 2280

(D) Rs. 2400

Correct Answer: (B) Rs. 2160

Solution:
Step 1: Calculate the daily wage for each worker.

Arun: $\frac{2160}{24} = 90$ Rs/day.

Tarun: $\frac{2400}{21} = \frac{800}{7} \approx 114.3$ Rs/day.

Varun: $\frac{2160}{15} = 144$ Rs/day.

Step 2: Calculate the efficiency (work per day) of each worker.

Let the Total Work be the LCM of the days: LCM(24, 21, 15) = 840 units.

Efficiency of Arun $(E_A) = \frac{840}{24} = 35$ units/day.

Efficiency of Tarun $(E_T) = \frac{840}{21} = 40$ units/day.

Efficiency of Varun $(E_V) = \frac{840}{15} = 56$ units/day.

Step 3: Calculate the Cost Efficiency (Cost per unit of work).

Cost per unit for Arun $= \frac{90}{35} = \frac{18}{7} \approx 2.57$ Rs/unit.

Cost per unit for Tarun $= \frac{800/7}{40} = \frac{20}{7} \approx 2.86$ Rs/unit.

Cost per unit for Varun $= \frac{144}{56} = \frac{18}{7} \approx 2.57$ Rs/unit.

Since Arun and Varun are equally cost-efficient and cheaper than Tarun, we should maximize their usage and avoid using Tarun unless the time constraint requires it.

Step 4: Check if the work can be completed in 10 days using only Arun and Varun.

Max work by Arun and Varun in 10 days $= 10 \times (35 + 56) = 10 \times 91 = 910$ units.

Since 910 units ¿ 840 units (Total Work), we do not need Tarun.

Step 5: Find the optimal number of days to minimize cost.

We need to choose integers d_A and d_V (days for Arun and Varun, ≤ 10) such that Total Work ≥ 840 and Cost is minimized.

Since Cost is proportional to Work for these two ($Cost = \frac{18}{7} \times Work$), we minimize Cost by finding a combination that yields exactly 840 units (or as close as possible above it).

Equation: $35d_A + 56d_V = 840$.

Divide by 7: $5d_A + 8d_V = 120$.

Let's test $d_V = 10$ (Max allowed for Varun):

$5d_A + 8(10) = 120 \implies 5d_A = 40 \implies d_A = 8$.

This gives exactly 840 units with valid integer days ($d_A = 8, d_V = 10$).

Step 6: Calculate the final cost.

Cost $= (8 \text{ days} \times 90) + (10 \text{ days} \times 144)$.

Cost $= 720 + 1440 = 2160$.

> Quick Tip
>
> First, determine the "Cost per Unit of Work" for each worker. Prioritize the most cost-efficient workers. Only consider less efficient workers if the time constraint cannot be met by the efficient ones.

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10. The number of non-negative values of n for which $\log_{1/4}(n^2 - 7n + 14) > 0$ **is**

(A) 0

(B) 1

(C) 2

(D) 4

Correct Answer: (A) 0

Solution:
Step 1: Understand the logarithmic inequality.

The inequality is given by $\log_{1/4}(n^2 - 7n + 14) > 0$.

Since the base of the logarithm is $\frac{1}{4}$ (which is between 0 and 1), the inequality sign reverses when converting to exponential form.

Also, the argument of the logarithm must be positive (Domain condition).

So, we require: $0 < n^2 - 7n + 14 < (1/4)^0$.

$0 < n^2 - 7n + 14 < 1$.

Step 2: Analyze the quadratic expression.

Let $f(n) = n^2 - 7n + 14$.

This is a parabola opening upwards (coefficient of n^2 is positive).

The minimum value occurs at the vertex, where $n = \frac{-b}{2a} = \frac{-(-7)}{2(1)} = 3.5$.

Calculate the minimum value of the expression:

$f(3.5) = (3.5)^2 - 7(3.5) + 14$.

$f(3.5) = 12.25 - 24.5 + 14$.

$f(3.5) = 1.75$.

Step 3: Check the conditions.

We need $f(n) < 1$.

However, we found that the minimum value of $f(n)$ is 1.75.

Since $f(n) \geq 1.75$ for all real values of n, it is never less than 1.

Therefore, the condition $n^2 - 7n + 14 < 1$ has no solution.

Conclusion:

There are no values of n (non-negative or otherwise) that satisfy the inequality. The number of such values is 0.

Quick Tip

For logarithmic inequalities $\log_b(x) > k$: if $0 < b < 1$, this converts to $0 < x < b^k$. Always check the minimum or maximum value of the argument x to see if the solution set is feasible.

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11. A container contains only milk. 2/3 of the mixture is removed and replaced with water. This process is done once and then repeated another 3 times. What is the final ratio of milk and water?

(A) 1 : 26

(B) 1 : 80

(C) 16 : 65

(D) 1 : 242

Correct Answer: (B) 1 : 80

Solution:
Step 1: Determine the fraction of milk remaining after one operation.

Let the initial total volume be V. Initially, Milk $= V$.

In each process, $\frac{2}{3}$ of the mixture is removed and replaced with water.

Fraction of milk remaining after one operation $= 1 - \frac{2}{3} = \frac{1}{3}$.

Step 2: Determine the total number of operations.

The process is done once first.

Then, it is repeated "another 3 times".

Total number of operations $(n) = 1 + 3 = 4$.

Step 3: Calculate the final quantity of milk.

Using the formula for repeated dilution:

Final Milk = Initial Milk $\times$ (Fraction Remaining)^n

Final Milk $= V \times \left(\frac{1}{3}\right)^4$

Final Milk $= V \times \frac{1}{81} = \frac{V}{81}$.

Step 4: Calculate the final quantity of water.

Since the total volume V remains constant (amount removed = amount replaced):

Final Water = Total Volume − Final Milk

Final Water $= V - \frac{V}{81} = \frac{80V}{81}$.

Step 5: Calculate the ratio.

Ratio of Milk : Water $= \frac{V}{81} : \frac{80V}{81}$.

Ratio $= 1 : 80$.

Quick Tip

Pay close attention to the phrasing of the repetition. "Repeated 3 times" usually implies $n = 3$, but "Repeated *another* 3 times" means $n = 1 + 3 = 4$.

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12. If the length of each side of a rhombus is 36 cm, and the area of the rhombus is 396 cm^2, then what is the absolute value of the difference between the lengths of its diagonals?

(A) 54 cm

(B) 60 cm

(C) 66 cm

(D) 72 cm

Correct Answer: (B) 60 cm

Solution:
Step 1: Use the area formula to find the product of the diagonals.

Let the lengths of the diagonals be d_1 and d_2.

The area of a rhombus is given by $A = \frac{1}{2}d_1 d_2$.

Given $A = 396$ cm^2.

$396 = \frac{1}{2}d_1 d_2 \implies d_1 d_2 = 396 \times 2 = 792.$

Step 2: Use the side length property to find the sum of the squares of the diagonals.

The diagonals of a rhombus bisect each other at right angles. This forms four right-angled triangles with legs $\frac{d_1}{2}$ and $\frac{d_2}{2}$ and hypotenuse equal to the side length s.

Using the Pythagorean theorem: $\left(\frac{d_1}{2}\right)^2 + \left(\frac{d_2}{2}\right)^2 = s^2$.

Multiply by 4: $d_1^2 + d_2^2 = 4s^2$.

Given side length $s = 36$ cm.

$d_1^2 + d_2^2 = 4 \times (36)^2 = 4 \times 1296 = 5184.$

Step 3: Calculate the difference between the diagonals using algebraic identities.

We need to find $|d_1 - d_2|$.

Recall the identity: $(d_1 - d_2)^2 = d_1^2 + d_2^2 - 2d_1d_2$.

Substitute the values calculated in steps 1 and 2:

$(d_1 - d_2)^2 = 5184 - 2(792)$.

$(d_1 - d_2)^2 = 5184 - 1584$.

$(d_1 - d_2)^2 = 3600$.

Step 4: Take the square root.

$|d_1 - d_2| = \sqrt{3600} = 60$.

Conclusion:

The absolute value of the difference between the lengths of the diagonals is 60 cm.

Quick Tip

For any rhombus with side s and diagonals d_1, d_2, remember the relation $d_1^2 + d_2^2 = 4s^2$. Combining this with the area formula ($2A = d_1d_2$) allows you to easily find sums ($d_1 + d_2$) or differences ($d_1 - d_2$) using algebraic identities.

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13. In a cafeteria, there are 5 breads. One can choose 1 bread from the available breads, either small or large sized, and can choose up to 2 sauces from 6 available sauces. What is the number of different ways one can place an order?

(A) 110

(B) 220

(C) 240

(D) 280

Correct Answer: (B) 220

Solution:
Step 1: Calculate the number of ways to choose the bread.

There are 5 types of breads available, and the customer must choose exactly 1.

Number of ways to choose bread $= \binom{5}{1} = 5$.

Step 2: Calculate the number of ways to choose the size.

For the chosen bread, there are 2 sizes available (Small or Large).

Number of ways to choose size $= \binom{2}{1} = 2$.

Step 3: Calculate the number of ways to choose the sauces.

The customer can choose "up to 2 sauces" from 6 available sauces. This means they can choose 0, 1, or 2 sauces.

Case 1: Choose 0 sauces.

Ways $= \binom{6}{0} = 1$.

Case 2: Choose 1 sauce.

Ways $= \binom{6}{1} = 6$.

Case 3: Choose 2 distinct sauces.

Ways $= \binom{6}{2} = \frac{6\times 5}{2} = 15$.

Total ways to choose sauces $= 1 + 6 + 15 = 22$.

Step 4: Calculate the total number of distinct orders.

Since the choices are independent events, we multiply the number of ways for each step.

Total Ways = (Ways for Bread) $\times$ (Ways for Size) $\times$ (Ways for Sauces).

Total Ways $= 5 \times 2 \times 22$.

Total Ways $= 10 \times 22 = 220$.

Quick Tip

When a constraint says "up to k items", calculate the sum of combinations for every possibility from 0 to k (i.e., $\binom{n}{0} + \binom{n}{1} + \cdots + \binom{n}{k}$).
